PortFlow Container Terminal: Yard and Gate Flow

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Recommendation. Option B — Dynamic yard: demand zoning, MIP/CRAFT slotting, tracking database, AI AGV pilot — delivers 21.7% lower year-5 yard utilization (target 19.5%) for CAD $9,400,000 of the CAD $10,000,000 budget: NPV CAD $6,091,191 at 9.5%, payback 32.7 months. Five-year net financial impact (case formula): -CAD $16,100,000 (net savings).

Capex	NPV @ WACC	IRR	Payback	Benefit/ Cost	Run-rate benefit	Levelized cost
CAD $9,400,000	CAD $6,091,191	28.1%	32.7 months	1.42	CAD $6,400,000/ yr	134474.76 CAD per year per % point of improvement

1. Problem diagnosis

The yard is static and blind. Today 21,120 of 24,000 TEU are in use (88%), containers stay 6 days and trucks take 72 minutes. With +20% volume the yard passes its capacity by year 5 unless containers are placed, moved and released faster; the 15-acre parcel alone would still leave about 90%.

- Core issue (cause): the yard runs on fixed rules with incomplete data. Blocks are not zoned by demand, stacks ignore pickup order and exact position, the layout never adapts as demand shifts by season, and release status is not shared.
- Wrong place: busy container groups sit far from the gate, rail and quay exits, so every move travels further (32 minutes inside).
- Wrong order: stacks are built without knowing pickup time or exact slot, so cranes reshuffle boxes to dig out the one needed (rehandles and $6M overtime).
- Wrong time: demand shifts (a product peaking in December but not in March) while the layout stays fixed, and customers learn late that boxes are released (6-day dwell, 40-minute queues).
- Symptoms, not causes: yard crowding, $6M overtime, $8M penalties, customers unable to plan and staff anxiety.

2. Options and decision

Opt	Alternative	Capex	Gain	In budget	Meets target	NPV	Payback	Score /10
A	Static re-zoning only: one-off MIP and CRAFT layout	CAD $1,600,000	3.9%	yes	no	CAD $4,032,937	20.0 months	5.22
B ✔	Dynamic yard: demand zoning, MIP/CRAFT slotting, tracking database, AI AGV pilot	CAD $9,400,000	21.7%	yes	yes	CAD $6,091,191	32.7 months	6.92
C	Yard expansion on the 15-acre parcel with two added RTGs	CAD $14,500,000	14.3%	NO	no	-CAD $14,627,035	> 60 months	2.9

Criteria and weights: Target attainment 30%, Financial return (NPV) 25%, Payback speed 15%, Downside resilience (-15% perf.) 15%, Budget headroom 15%. Budget is a hard gate.

3. Five-year cash flow (CAD)

Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
-CAD $9,400,000	CAD $620,000	CAD $5,100,000	CAD $5,100,000	CAD $5,100,000	CAD $5,100,000

4. Scenarios

Scenario	Gain	Capex	NPV	IRR	Payback	Meets target
Bear (-15% perf., +20% capex)	18.45%	CAD $11,280,000	CAD $140,445	9.9%	46.7 months	no
Base (approved plan)	21.7%	CAD $9,400,000	CAD $6,091,191	28.1%	32.7 months	yes
Bull (+15% perf., -10% capex)	24.95%	CAD $8,460,000	CAD $10,602,775	42.6%	26.4 months	yes

5. Sensitivity (NPV tornado)

Driver	Low NPV	High NPV	Swing
Annual benefit: achieved gain x unit value (+/-20%)	CAD $1,994,628	CAD $10,187,753	CAD $8,193,125
O&M cost (x2 / x0.5)	CAD $1,099,569	CAD $8,587,001	CAD $7,487,432
Capital cost (+20% / -20%)	CAD $3,212,866	CAD $8,969,515	CAD $5,756,649
Implementation duration (+50% / -25%)	CAD $4,223,006	CAD $7,114,022	CAD $2,891,016

6. Implementation roadmap

Horizon	Timing	Workstream	Budget	KPI	Gate
Phase 1	Months 1-3	Design: data, layout model, AGV safety case	CAD $2,200,000	Move data cleaned; MIP/CRAFT layout validated in simulation; AGV vendor and safety case approved	Layout model beats today's travel by >= 20% in simulation; AGV safety case signed
Phase 2	Months 4-9	Re-zone, go dynamic, pilot 2 AGVs	CAD $5,600,000	Dwell <= 5.3 days; turn <= 55 min; rehandles -20%; AGVs commissioned	Slotting advice accepted >= 70%; AGV availability >= 90% in the fenced lane
Phase 3	Months 10-12	Scale decision and hand over	CAD $1,600,000	Dwell <= 4.8 days; turn < 45 min; utilization on track for <= 85%	AGV cost per move <= manual tractor before any fleet expansion; savings confirmed

7. Risk register

Risk	L	I	Mitigation	Trigger
AGV breakdown or navigation fault	Medium	High	Preventive maintenance plan and vendor SLA, sensor health monitoring, fenced lane, manual tractors as instant fallback	AGV availability < 90% or any unplanned stop in the lane
Tracking or database errors (wrong slot recorded)	Medium	High	OCR and crane-sensor double check on every move, nightly reconciliation audit, manual search procedure kept	Position accuracy < 99% in the daily audit
AI slotting or demand forecast wrong	Medium	Medium	Re-slots run weekly on actual bookings; planners approve; segregation rules locked; 6 weeks of shadow mode	Override rate > 30% or rehandles rise for 2 weeks
Re-zoning moves disrupt live operations	Medium	High	One block at a time, overnight housekeeping moves only, go-lives outside vessel peaks, rollback plan	Gate throughput below baseline for 2 shifts
Employee resistance to AGVs and AI	Medium	Medium	No layoffs from this program; drivers retrain as AGV supervisors and technicians; staff co-design before tools arrive	Training completion < 90% or advice acceptance < 50%
AGV pilot costs more than it saves	Medium	Medium	Fixed-price quote, pilot capped at 2 units, fleet expands only if the month-12 gate passes	Cost per move above manual tractor after 3 months

8. Judge defence

Q: Why not just build on the 15-acre parcel?
Space alone fails the objective: 25,344 / 28,000 TEU = 90.5% at year 5, for about $14.5M, over budget. Cutting dwell to 4.7 days reaches about 83% on today's footprint.

Q: Why only two AGVs?
The evidence is mixed: successful automated terminals cut operating costs 25-55%, but some lose 7-15% productivity. Two units in a fenced lane prove uptime, cost per move and safety for $1.6M; the fleet grows only if the month-12 gate passes.

Q: How do the MIP and CRAFT work together?
The MIP assigns container groups to blocks, minimising move frequency x distance to their exit within capacity and segregation rules. CRAFT then tries pairwise zone swaps and keeps any that lower total flow x distance, so the layout can be re-improved as demand shifts.

Q: What if the seasonal forecast is wrong?
Slotting re-runs weekly on actual bookings, not only the forecast; planners approve every re-slot; zoning is reviewed monthly.

Q: How do you know where every container is?
OCR cameras read container numbers at the gate and quay, crane position sensors record the exact slot of every pick and drop, and RFID tags identify each AGV and truck. Every move writes to one yard database that the slotting

engine, the AGVs and the customer portal all read.

Q: Will automation cost jobs, and who maintains the AGVs?
No layoffs from this program. Drivers can retrain as AGV supervisors and technicians. AGVs follow a preventive maintenance plan with a vendor SLA, and manual tractors stay as the fallback.

9. Assumptions

Parameter	Value	Source
Baseline	105.6 % of capacity	case datum
Target improvement	19.5%	case datum
Capital budget	CAD $10,000,000	case datum
Timeline	12 months	case datum
Option costs & gains	see options table	case datum
Monetisation	option-level annual savings stated in the case pack	assumption: estimate (overtime -40%, penalties -50%; dwell 6.0 to 4.7 days)
Discount rate (WACC)	9.5%	assumption (mid-cap industrial hurdle)
Appraisal horizon	5 years	assumption
Annual O&M	CAD $1,300,000/yr (14% of capex)	assumption: estimate (overtime -40%, penalties -50%; dwell 6.0 to 4.7 days)
Benefit during implementation	30% of full run-rate	assumption (pilot ramp)